Fractalia: Deep-Zoom Rendering of Composable Fractals

Mathematical specification, implementation, and arithmetic cost

Abstract

Fractalia is a browser renderer for 24 orbit primitives, their cyclic compositions, and periodically forced logistic maps. This report specifies the maps and their observables, derives the differences used to share high-precision reference orbits, and describes the conditions for accepting an approximation or evaluating a pixel directly. An operation-count model covers all 52 presets and one mixed Warp recipe. Between magnifications 10^{12} and 10^{100}, it predicts an 8.18-fold increase in direct arithmetic work for binary fixed-point kernels with a fixed operation mix, and a 15.45-fold increase for Warp because its residual sine polynomial also gains terms. Archived frame timings show why these ratios do not predict latency across regions or rendering paths. We distinguish established methods, engine-specific engineering and measurements, and candidates for further study. The evidence supports an implementation and evaluation contribution; it does not establish a new general perturbation method or priority for the individual identities.

Keywords: fractal rendering, perturbation, fixed-point arithmetic, hybrid maps, operation counts, deep zoom.

Contents

1 Introduction and scope 2

2 Mathematical specification 2

3 Formula Studio: construction and specialization 5

4 Precision and routing 6

5 Numerical methods and derivations 7

6 Arithmetic work model 14

7 Results and reproducibility 18

8 Prior art and contribution assessment 21

References 22

A Results for every preset 24

1 Introduction and scope

A deep view may place adjacent pixels much closer together than one floating-point unit at the viewport center. Evaluating the entire image with large integers preserves the distinction but can make ordinary interaction slow. Reference-orbit perturbation separates the cost of storing a deep location from the cost of tracking nearby orbits. Martin's technical note and Khashin's preprint describe this idea for Mandelbrot rendering; Heiland-Allen documents rebasing, hybrid maps, and linear approximation [7, 8, 9, 10].

Fractalia's design question is narrower: how can a finite library of heterogeneous steps share this rendering machinery while retaining its own initial conditions, convergence rules, orbit traps, and fallback? The report specifies the implementation and gives a reproducible arithmetic inventory. It is a technical report on the local checkout, including the optimizations beyond revision `247a269`. The work model is analytical; the archived timings are descriptive. The novelty and strength of evidence are assessed in Section 8.

2 Mathematical specification

2.1 Sampling and finite observables

Let W, H be raster dimensions, C the complex viewport center, and $s > 0$ its width in the complex plane. For integer indices i, j the renderer samples the centers of the final raster cells:

$$p_{ij} = C + \frac{s}{W}\left(i + \frac{1}{2} - \frac{W}{2} + \mathrm{i}\left(j + \frac{1}{2} - \frac{H}{2}\right)\right), \tag{2.1}$$

$$Z_{\mathrm{zoom}} = \frac{3.3}{s}, \qquad x = \log_{10} Z_{\mathrm{zoom}}. \tag{2.2}$$

The imaginary coordinate increases down the screen. The coarse passes sample at stride h, replacing $i + \frac{1}{2}$ and $j + \frac{1}{2}$ by $h(i + \frac{1}{2})$ and $h(j + \frac{1}{2})$. Pan can express the same lattice using a retained anchor and integer shifts. The implementation is in `worker.js` and `math.js`.

A parameter plane varies c for a chosen initial state; a dynamical plane varies the initial state for fixed c. Julia and Mandelbrot give the historical context [1, 2]. The image represents the chosen finite-budget observable. In particular, consuming N iterations does not prove membership in a bounded set, and a bailout chosen for a hybrid, rational, or transcendental map need not be a mathematical escape radius.

2.2 Orbit primitives

Write $z = a + \mathrm{i}b$, $u = a^2 - b^2$, $v = 2ab$, and $c = c_n(p)$. Table 1 gives all 24 identifiers grouped by shared expressions. The implementation sources are `formulas.js` and `precise-formulas.js`.

Table 1: The orbit primitives before bailout or convergence tests.

Identifier	z_{n+1}	Interpretation
`z2`, `z3`, `z4`, `z5`, `z6`, `z7`	$z^k + c$, $k = 2, \ldots, 7$	Quadratic Mandelbrot or Julia; higher-degree Multibrot or Julia.

Identifier	z_{n+1}	Interpretation
`tricorn`, `multi3`, `multi4`, `multi5`	$\overline{z}^{\,k} + c$, $k = 2, 3, 4, 5$	Conjugate before taking the power.
`ship`	$(\lvert a\rvert + \mathrm{i}\lvert b\rvert)^2 + c$	Burning Ship [3].
`perpendicular`	$(\lvert a\rvert - \mathrm{i}b)^2 + c$	Fold real, reflect imaginary.
`heart`	$(\lvert a\rvert + \mathrm{i}b)^2 + c$	Fold real before squaring.
`celtic`	$\lvert u\rvert + \mathrm{i}v + c$	Fold the real component of the square.
`buffalo`	$\lvert u\rvert - \mathrm{i}\lvert v\rvert + c$	Fold both components of the square.
`warp`	$z^2 + 0.35[\sin(3a) + \mathrm{i}\sin(3b)] + c$	Two real sines; not a complex sine.
`phoenix`	$z^2 + mz_{n-1} + c$	Real memory weight m; $z_{-1} = 0$.
`lambda`	$c(z - z^2)$	The parameter is a multiplier.
`cubic`	$z^3 - 0.9z + c$	Critical points $\pm\sqrt{0.3}$.
`newton`	$(2z + \theta_n/z^2)/3$	$\theta_n = 1 + c$ with Julia; $1 + c - p$ otherwise.
`nova`	$(2z + 1/z^2)/3 + c$	Add the parameter after the Newton update.
`magnet`	$\big((z^2 + c - 1)/(2z + c - 2)\big)^2$	Rational map with a test near 1.
`exp`	$e^z + c$	Complex exponential.
`sine`	$\sin z + c$	Complex sine.

Newton and Nova. At one step, the root target θ is fixed. For $F(z) = z^3 - \theta$ and $z \neq 0$, Newton's rule [4] gives

$$\begin{aligned} F'(z) &= 3z^2, \\ z - \frac{F(z)}{F'(z)} &= z - \frac{z^3 - \theta}{3z^2} \\ &= \frac{2z + \theta/z^2}{3}. \end{aligned}$$

The unmodified Newton preset uses $\theta = 1$. Nova applies the same update with $\theta = 1$, then adds c. Its color sector is an observable, not a verified root of $z^3 - 1$ when $c \neq 0$.

Cubic and transcendental maps. The critical seeds are obtained from

$$\begin{aligned} \frac{d}{dz}(z^3 - 0.9z + c) &= 3z^2 - 0.9, \\ 3z^2 - 0.9 = 0 \iff z &= \pm\sqrt{0.3}. \end{aligned}$$

The preset with two seeds returns the smaller scalar observable of the two orbits. Sine and exponential are evaluated through real functions [17]:

$$e^{a+\mathrm{i}b} = e^a(\cos b + \mathrm{i}\sin b), \tag{2.3}$$

$$\sin(a + \mathrm{i}b) = \sin a \cosh b + \mathrm{i}\cos a \sinh b. \tag{2.4}$$

2.3 Termination, traps, and color

The default escape radius is 16; a single Newton step uses 10^{14}. For a degree-k polynomial well outside the bounded region, the motivation for smooth dwell [5] is

$$|z_{n+1}| \approx |z_n|^k,$$
$$\log|z_{n+1}| \approx k \log|z_n|,$$
$$\log_k(\log|z_{n+1}|) \approx 1 + \log_k(\log|z_n|).$$

Fractalia uses, at the first escape after step n,

$$V = n + 1 - \log_k(\log_2|z_{n+1}|). \tag{2.5}$$

Here k is the designated exponent of the escaping step, defaulting to 2 for steps without an explicit power, including Cubic, Warp, Sine and Exponential. Without an escape, the escape observable is N.

For single-step Newton and Nova, convergence is tested before the escape and trap update; Magnet tests proximity to 1. Their scalar rules are

$$|z_{n+1} - z_n|^2 < 10^{-12} \implies V = 3 + 14B + \min(10, 2\log_2(n+1)), \tag{2.6}$$
$$B = 1 + \operatorname{round}\left(\frac{3\arg(z_{n+1})}{2\pi}\right), \tag{2.7}$$
$$|z_{n+1} - 1|^2 < 10^{-12} \implies V = 25 + \min(20, 0.2n) \quad \text{(Magnet)}. \tag{2.8}$$

These single-step convergence tests are disabled in longer sequences. At a zero denominator the direct evaluator returns N; extreme transcendental arguments can trigger a numerical guard before evaluating the function. With a trap, a converging step returns the previously accumulated trap value.

For trap position ρ, define

$$d_{\text{ring}}(z) = \big||z| - \rho\big|, \tag{2.9}$$
$$d_{\text{cross}}(z) = \min\{|\Re z - \rho|, |\Im z - \rho|\}, \tag{2.10}$$
$$d_* = \min_n d(z_{n+1}), \tag{2.11}$$
$$V_{\text{trap}}(d_*) = \min\{N - 0.001, \max(0, -6\log_2(d_* + 10^{-12}))\}. \tag{2.12}$$

Trap modes change the observable, not the orbit. The cached scalar field is mapped to RGBA separately. The phase samples the palette at $\operatorname{frac}(0.021V + \phi/360)$, using $f^2(3-2f)$ between adjacent colors. The background is $(10, 14, 25)$. Optional glow blends the palette toward this background with

$$h = \operatorname{clamp}_{[0,1]}\big((V - \texttt{glowStart})/2.25\big),$$
$$w = 0.065 + 0.64h^2(3 - 2h).$$

See `colorize.js`.

2.4 Periodically forced Lyapunov mode

The pixel is the pair $(A, B) \in [0, 4]^2$. A periodic binary word selects $r_n \in \{A, B\}$. If $f_r(u) = ru(1-u)$, then

$$u_0 = \tfrac{1}{2}, \qquad u_{n+1} = f_{r_n}(u_n), \tag{2.13}$$

$$f'_{r_n}(u_n) = r_n(1 - 2u_n), \tag{2.14}$$

$$\frac{\partial u_{n+1}}{\partial u_0} = \prod_{j=0}^{n} f'_{r_j}(u_j). \tag{2.15}$$

The last line follows by the chain rule. Taking the logarithm of the absolute derivative turns a product into a sum. With 24 discarded iterations and a floor for zero derivatives, the displayed estimate is

$$\widehat{\lambda}_N = \frac{1}{N} \sum_{n=24}^{N+23} \log \max\{10^{-15}, |r_n(1 - 2u_n)|\}. \tag{2.16}$$

Outside $[0, 4]^2$ the value is N. Negative estimates use the trap color function with distance $-\widehat{\lambda}_N$; nonnegative ones use $34 + \min(14, 22\widehat{\lambda}_N)$. The preset words are `AABAB` and `ABBABAAB`. The forcing and Lyapunov images are established prior art [6]; the finite estimate does not classify the asymptotic orbit with certainty.

3 Formula Studio: construction and specialization

3.1 Recipe semantics

A recipe is normalized data, not executable input. It holds one to eight steps, each identified by one of the 24 primitives and repeated an integer $R_j \in [1, 16]$ times. Per-step controls include Julia, a complex constant $J_j \in [-8, 8]^2$, and Drift with amplitude $A_j \in [0, 2]$ and period $P_j \in [2, 96]$. Global controls select the seed, escape/convergence or a trap, $\rho \in [0.01, 4]$, Phoenix memory $m \in [-2, 2]$, or a Lyapunov word of length 1–24. Missing or invalid inputs are normalized by `formulas.js`.

Let S be the ordered word obtained by repeating each step as specified, of length $L = \sum_j R_j \le 128$. At global iteration n the step is $j = S[n \bmod L]$. The parameter is

$$c_j^{(0)}(p) = \begin{cases} J_j, & \text{Julia is enabled for step } j, \\ p, & \text{otherwise,} \end{cases} \tag{3.1}$$

$$d_j(n) = \begin{cases} A_j \exp(2\pi \mathrm{i}(n \bmod P_j)/P_j), & \text{Drift is enabled,} \\ 0, & \text{otherwise,} \end{cases} \tag{3.2}$$

$$c_n(p) = c_j^{(0)}(p) + d_j(n). \tag{3.3}$$

The forcing period divides $\operatorname{lcm}(L, P_{j_1}, \ldots, P_{j_k})$ over active nonzero drift steps. Indeed, shifting n by that least common multiple preserves both the recipe index and every active drift phase. The state itself need not repeat. Phoenix retains z_{n-1} at every step of a mixed recipe, including steps that do not read the memory.

Automatic seeding uses the pixel for an initial Julia step or Newton. Otherwise Lambda uses 1/2, Nova uses 1, Cubic uses its critical seed, and other steps start at zero. Explicit seeds are Pixel, Zero and Formula seed; Formula seed uses 1 outside the Lambda/Cubic cases. Two seeds are evaluated only for a single Cubic step with a critical, nonpixel seed.

Running example. The study recipe consists of a cubic Julia step with Drift, followed by three parameter-plane Warp steps. For $n \bmod 4 = 0$,

$$\begin{aligned} z_0 &= p, \\ c_n &= -0.23 + 0.53\mathrm{i} + 0.08\exp(2\pi\mathrm{i}(n \bmod 13)/13), \\ z_{n+1} &= z_n^3 + c_n. \end{aligned}$$

For the other residues $c_n = p$ and the Warp map applies. The combined forcing period is $\mathrm{lcm}(4, 13) = 52$.

3.2 Generation and rendering

`Shuffle` samples up to 16 candidate recipes. Each contains two to four steps drawn with replacement from the first steps of the Hybrid and Drift presets, with one to three repetitions per step. A 24×16 preview at $N = 200$ accepts candidates with 12–288 budget-limited points and at least ten distinct floored escape values. Otherwise it uses one of three known recipes. This is a visibility heuristic, not a classification of the dynamics. See `catalog.js` and `studio.js`.

```
Normalize the recipe and determine the seed(s).
If it matches a classic map, route to the classic evaluator.
Otherwise prepare the Float64 or fixed-point step descriptors.
Cache the per-step Drift tables.
For a supported deep view, trace high-precision reference orbits.
Attach per-step power, fold, derivative and quotient data.
Use an eligible shared prefix, then propagate the pixel differences.
Retry with another reference if uncertain; otherwise evaluate directly.
```

Classic routing recognizes a single escape step without Drift: quadratic Julia with pixel seed, or quadratic Mandelbrot or Burning Ship with zero seed and no Julia parameter. A non-drifting quadratic trap has its own shallow evaluator. At depth, general Studio recipes share a kernel whose metadata and recurrences depend on the ordered steps. Single-step Sine and Exponential Julia may instead use a specialized analytic evaluator. The engine builds descriptors and evaluators from a fixed vocabulary. It neither emits new JavaScript source nor derives arbitrary user expressions symbolically.

4 Precision and routing

The view stores the center and span as decimal-scaled integers with scale $S_d = 10^d$. A fresh view starts at $d = 40$. If the integer span has fewer than 25 decimal digits, the state multiplies its integers by 10^{20} and increments d by 20. This preserves the represented point but cannot restore previously discarded digits. The classic direct evaluator rescales each real component of the complex square by integer division by S_d.

Studio and the Burning Ship reference path use binary fixed point:

$$b(x, W) = \lceil (x + \log_{10} W + 32) \log_2 10 \rceil , \tag{4.1}$$

$$U = 2^b, \tag{4.2}$$

$$\mathrm{mul}_b(A, B) = \lfloor AB/U \rfloor , \tag{4.3}$$

$$\mathrm{div}_b(A, B) = \mathrm{trunc}(UA/B). \tag{4.4}$$

The $x+\log_{10} W$ term follows pixel spacing up to the constant $\log_{10} 3.3$, and 32 adds decimal guard digits. A right shift implements multiplication rescaling; signed integer division truncates toward zero. Columns and rows share the coordinate conversions. The word *direct* means finite-precision evaluation without perturbation, not exact real arithmetic.

The main deep-zoom condition is $x > 11$. A view entered through Studio can also switch when

$$s/W \le 128\, 2^{-52} \max\{1, |\Re C|, |\Im C|\}. \tag{4.5}$$

Mandelbrot and Julia then use decimal references. Burning Ship uses binary reference traces and a decimal direct fallback; other Studio recipes use binary traces and binary fallback. The reference routes require $s > 10^{-250}$; below this threshold a direct path remains available, including an eligible fixed-point block test. This is a routing threshold, not a maximum zoom. Practical limits include memory, latency, finite coordinate inputs and the app's input caps. The analytic path requires a single non-drifting Sine or Exponential Julia step, escape coloring, and Auto or Pixel seed.

5 Numerical methods and derivations

5.1 Quadratic perturbation and polynomial prefixes

Let Z_n be the reference state and C its parameter. Represent the pixel by the differences $\delta_n = z_n - Z_n$ and $\eta = c - C$. For the quadratic map, the two orbits give

$$Z_{n+1} = Z_n^2 + C, \tag{5.1}$$

$$z_{n+1} = (Z_n + \delta_n)^2 + C + \eta, \tag{5.2}$$

$$\begin{aligned} \delta_{n+1} &= z_{n+1} - Z_{n+1} \\ &= 2Z_n\delta_n + \delta_n^2 + \eta. \end{aligned} \tag{5.3}$$

The last line avoids subtracting rounded, nearly equal orbit values. Mandelbrot starts with $\delta_0 = 0$; classic Julia has $\eta = 0$ and starts from a displaced seed. This construction and prefix skipping are established [7, 8, 9].

For the shared Mandelbrot prefix, write $\eta = Ru$, with $|u| \le 1$, and keep four powers of u:

$$P_n(u) = \sum_{k=1}^{4} a_{n,k} u^k, \qquad a_{0,k} = 0, \tag{5.4}$$

$$a_{n+1,k} = 2Z_n a_{n,k} + \sum_{\substack{i,j\ge 1 \\ i+j=k}} a_{n,i} a_{n,j} + R\,\mathbf{1}_{k=1}. \tag{5.5}$$

Equation (5.5) follows by collecting coefficients in $2Z_nP_n + P_n^2 + Ru$. To bound the residual, let $e_n = \delta_n - P_n$, $B_n = \sum_k |a_{n,k}|$, and $|e_n| \le E_n$. The discarded terms of P_n^2 have total magnitude at most $T_n = \sum_{i+j>4} |a_{n,i}||a_{n,j}|$. Subtracting the retained polynomial gives

$$e_{n+1} = (2Z_n + 2P_n)e_n + e_n^2 + [P_n^2]_{\deg>4}, \tag{5.6}$$

$$E_{n+1} \le (2|Z_n| + 2B_n + E_n)E_n + T_n \quad \text{in exact arithmetic.} \tag{5.7}$$

The implementation adds a rounding allowance to the right-hand side. It keeps advancing only if the estimated error is at most $10^{-12}B_{n+1}$, $B_{n+1} + E_{n+1} < 0.01$, and the enclosing disk stays inside radius 16. A prefix of at least eight steps is evaluated by Horner's rule; the pixel resumes the full recurrence afterwards.

Rebasing and recovery. If the reconstructed orbit becomes smaller than its difference, or the reference ends, Mandelbrot can rebase the current state to $Z_0 = 0$ without resetting the pixel iteration or parameter. Severe cancellation requests recovery. After rebasing, an estimated amplified error above 10^{-8} can trigger another reference near $(s/5, -s/7)$ from the center; agreement in class and within 10^{-4} in the scalar value can avoid a direct retry. The Julia path instead falls back for a short reference or when $|Z_n + \delta_n|^2 < 10^{-7}|Z_n|^2$.

For the Mandelbrot recovery route at pixel step above 10^{-26} and $|\Re C| + |\Im C| < 8$, each component is a pair of doubles $h + \ell$. A Dekker-style split with factor $2^{27} + 1$ captures the rounding residual of the leading product [16]. The retained product correction also includes $h_A\ell_B + \ell_A h_B$; a TwoSum-style residual corrects additions. BigInt is used when the propagated double-double error estimate exceeds 10^{-9}. See `mandelbrot-precision.js`.

Shallow analytic membership tests. For the main cardioid, an attracting fixed point satisfies $c = z - z^2$ and $|2z| < 1$. Put $c = a + \mathrm{i}b$, $w = c - \frac{1}{4}$, $u = \Re w$, $r = |w|$, and $q = r^2$. Choose $\sqrt{-w}$ with nonnegative real part. Then

$$\begin{aligned}
2z &= 1 - 2\sqrt{-w}, \\
\left(\Re\sqrt{-w}\right)^2 &= (r - u)/2, \\
|2z|^2 &= 1 - 4\sqrt{(r-u)/2} + 4r, \\
|2z| \le 1 &\iff 2q + u \le r \\
&\iff (2q + u)^2 \le q \quad (u \ge -r) \\
&\iff q(q + u) \le b^2/4.
\end{aligned}$$

For period two, $f(z) = z^2 + c$ satisfies

$$\begin{aligned}
f(f(z)) - z &= (z^2 - z + c)(z^2 + z + c + 1), \\
z_1 z_2 &= c + 1, \\
|f'(z_1)f'(z_2)| \le 1 &\iff |c + 1| \le \tfrac{1}{4}.
\end{aligned}$$

The two inequalities, with boundaries included, are the early tests used by the classic Float64 Mandelbrot path.

5.2 Differences for composed steps

The parameter difference η is zero on Julia steps and equals the pixel displacement on parameter-plane steps. The Drift offset in Equation (3.3) cancels between pixels at the same iteration. At Newton steps the root target is shared even in the parameter plane because p cancels in $1+c-p$.

Powers and folds. Put $D_k = (Z+\delta)^k - Z^k$. Factoring one power gives a recurrence without cancellation:

$$\begin{aligned} D_k &= (Z+\delta)(Z+\delta)^{k-1} - Z\,Z^{k-1} \\ &= (Z+\delta)\big[(Z+\delta)^{k-1} - Z^{k-1}\big] + \delta Z^{k-1} \\ &= (Z+\delta)D_{k-1} + \delta Z^{k-1}. \end{aligned} \tag{5.8}$$

For conjugation, conjugate Z and δ before the recurrence. For an absolute-value fold with real reference r and offset h, the exact difference can be evaluated without directly subtracting the two magnitudes:

$$|r+h| - |r| = \begin{cases} h, & r \ge 0,\ r+h \ge 0, \\ -2r-h, & r \ge 0,\ r+h < 0, \\ -h, & r < 0,\ r+h \le 0, \\ 2r+h, & r < 0,\ r+h > 0. \end{cases} \tag{5.9}$$

Ship, Perpendicular, and Heart use this on inputs; Celtic and Buffalo use it after squaring. Related folded perturbation is established [9, 10, 11].

Cubic, Phoenix, and Lambda. The updates, with $g = Z - Z^2$ and $\Delta g = \delta - D_2$, are

$$\Delta_{\text{Cubic}} = D_3 - 0.9\delta + \eta, \tag{5.10}$$

$$\Delta_{\text{Phoenix}} = D_2 + m\delta_{\text{prev}} + \eta, \tag{5.11}$$

$$\begin{aligned} \Delta_{\text{Lambda}} &= (C+\eta)(g+\Delta g) - Cg \\ &= C\Delta g + \eta(g+\Delta g). \end{aligned} \tag{5.12}$$

Cubic takes the magnitude *after* forming the derivative $3z^2 - 0.9$. Taking magnitudes first would lose cancellation at its critical points.

Warp, Sine, and Exponential. Use addition identities and cancellation-resistant forms of the small correction terms. For either real Warp component,

$$\begin{aligned} \sin(t+h) - \sin t &= \cos t \sin h + \sin t(\cos h - 1) \\ &= \cos t \sin h - 2\sin t \sin^2(h/2). \end{aligned} \tag{5.13}$$

Here t and h are three times the corresponding reference and displacement components. The two corrections, scaled by 0.35, augment $D_2 + \eta$. For the complex analytic functions,

$$e^{Z+\delta} - e^Z = e^Z(e^\delta - 1), \tag{5.14}$$

$$\sin(Z+\delta) - \sin Z = \cos Z \sin\delta + \sin Z(\cos\delta - 1). \tag{5.15}$$

For $\delta = u + \mathrm{i}v$, the residual parts are evaluated as

$$\Re(e^{\delta} - 1) = \operatorname{expm1}(u) - 2e^{u}\sin^2(v/2), \tag{5.16}$$

$$\Im(e^{\delta} - 1) = e^{u}\sin v, \tag{5.17}$$

$$\Re(\cos\delta - 1) = 2\big[\sinh^2(v/2) - \sin^2(u/2)\cosh v\big], \tag{5.18}$$

$$\Im(\cos\delta - 1) = -\sin u \sinh v. \tag{5.19}$$

The real-part formulas follow by adding and subtracting e^{u} or $\cosh v$, then using half-angle identities. The general Studio kernel finally adds η. These are algebraic constructions from standard identities [17], not new trigonometric results.

Lemma 5.1 (Difference of a quotient)**.** *If $Q = A/B$, $B \neq 0$, and $B + \Delta B \neq 0$, then*

$$\Delta Q = \frac{\Delta A - Q\Delta B}{B + \Delta B}. \tag{5.20}$$

If $|\Delta A| \leq r_A$ and $|\Delta B| \leq r_B < |B|$, then

$$|\Delta Q| \leq \frac{r_A + |Q|\, r_B}{|B| - r_B}. \tag{5.21}$$

Proof.

$$\begin{aligned}\frac{A + \Delta A}{B + \Delta B} - Q &= \frac{A + \Delta A - QB - Q\Delta B}{B + \Delta B} \\ &= \frac{\Delta A - Q\Delta B}{B + \Delta B}.\end{aligned}$$

The numerator is bounded above by $r_A + |Q|\, r_B$ by the triangle inequality. The reverse triangle inequality bounds the denominator below by $|B| - r_B > 0$. □

For Newton/Nova take $A = \theta$ (unit target for Nova), $B = Z^2$, $\Delta A = 0$, and $\Delta B = D_2$. The difference of the update is $(2\delta + \Delta Q)/3$, plus η for Nova. For Magnet use

$$A = Z^2 + C - 1, \qquad B = 2Z + C - 2, \tag{5.22}$$

$$\Delta A = D_2 + \eta, \qquad \Delta B = 2\delta + \eta, \tag{5.23}$$

$$\Delta_{\text{Magnet}} = 2Q\Delta Q + (\Delta Q)^2. \tag{5.24}$$

To tighten the error estimate near a reconstructed pixel, the engine uses derivatives in forms that retain cancellation. By the quotient rule,

$$\partial_z Q_{\text{Magnet}} = \frac{2zB - 2A}{B^2} = \frac{2(z - Q)}{B}, \tag{5.25}$$

$$\partial_c Q_{\text{Magnet}} = \frac{B - A}{B^2} = \frac{1 - Q}{B}, \tag{5.26}$$

$$\partial_z[(2z + Q_{\text{Newton}})/3] = \frac{2 - 2Q_{\text{Newton}}/z}{3} = \frac{2(z - Q_{\text{Newton}})}{3z}. \tag{5.27}$$

A denominator whose uncertainty interval approaches zero rejects the fast result. Local derivatives can tighten a bound but cannot remove a pole or resolve ambiguous convergence by themselves.

Logistic differences. Let U, R denote the scalar state and parameter of the reference, and let the pixel use $U + \delta, R + \eta$. Expanding before evaluating the small terms gives

$$\begin{aligned}\delta_{\text{next}} &= (R+\eta)(U+\delta)(1-U-\delta) - RU(1-U) \\ &= [R(1-2U) - R\delta]\delta + \eta(U+\delta)(1-U-\delta), \qquad (5.28)\\ \Delta f' &= (R+\eta)(1-2(U+\delta)) - R(1-2U) \\ &= -2R\delta + \eta(1-2(U+\delta)). \qquad (5.29)\end{aligned}$$

The fast kernel bounds the log-sum uncertainty and rejects estimates whose sign is ambiguous. For a positive lower derivative bound, $\log(1+t) \le t$ for $t \ge 0$ avoids another logarithm when bounding the error. The floor 10^{-15} used in the observable still applies.

5.3 Shared prefixes, delayed state, and fallback

General linear prefix. Let $u \in \mathbb{R}^2$ be the normalized pixel displacement in a reference tile. Write $\delta_n \simeq A_n u$, where A_n is a real 2×2 matrix. The chain rule gives the linearized response of a recipe:

$$A_{n+1} = J_{z,n} A_n + J_{c,n} L_n + m A_{n-1} \mathbf{1}_{\text{Phoenix}}. \qquad (5.30)$$

Here $L_n u$ is the parameter displacement, zero on Julia steps. The estimate accumulates the previous error, a second-order remainder $\frac{1}{2} M_n \|\delta\|^2$, the parameter–state cross term for Lambda, and rounding allowances. The code stops the prefix at an uncertain fold, an unsafe bailout margin, or excessive relative error (threshold 10^{-11} against the local scale). It requires at least 24 steps. Traps and rational steps disable this prefix because intermediate extrema, poles, or convergence can affect the observable.

Phoenix cancellation across steps. For a reconstructed state z_n, let ϵ_n denote a small state error. A Phoenix step gives

$$\epsilon_{n+1} = 2 z_n \epsilon_n + m \epsilon_{n-1} + \epsilon_n^2 + \text{numerical noise}, \qquad (5.31)$$

$$\begin{pmatrix} \epsilon_{n+1} \\ \epsilon_n \end{pmatrix} = \underbrace{\begin{pmatrix} 2z_n & m \\ 1 & 0 \end{pmatrix}}_{T_n} \begin{pmatrix} \epsilon_n \\ \epsilon_{n-1} \end{pmatrix} + \begin{pmatrix} \epsilon_n^2 + \text{numerical noise} \\ 0 \end{pmatrix}. \qquad (5.32)$$

Taking magnitudes before composing loses correlations. For two steps, put $a_j = 2 z_j$; the coefficient of the initial current error is

$$\begin{aligned}(T_1 T_0)_{11} &= a_1 a_0 + m, \\ |(T_1 T_0)_{11}| &\le |a_1 a_0| + |m|.\end{aligned}$$

For $a_0 = a_1 = 1/2$, $m = -1/4$, the left-hand side is zero while the triangle bound is $1/2$. This illustrates cancellation, not a benchmark. The implementation composes up to eight consecutive Phoenix steps before taking coefficient norms. It propagates nonlinear noise separately and keeps the smaller of its two estimates. A non-Phoenix step interrupts the block.

Reference policy. General Studio uses 64×64 tiles and a bounded per-worker cache. It can retry with a 32-pixel shifted grid, and, for selected formulas and sufficient cache budget, 16×16 subtiles. Intermediate error limits are 10^{-5} for most steps and 10^{-3} for rational steps. Bailout, trap color, convergence, and Lyapunov sign also receive decision-specific checks. When accepting a scalar interval, the general kernel uses its midpoint only if the width is at most 10^{-6}. After 64 consecutive tile failures (8 for Lyapunov) it temporarily bypasses the fast path for up to 256 pixels. For two seeds, both approximate values must be finite. These rules and their rounding allowances are engineering acceptance tests, not a directed-rounding certificate. See `studio-perturbation.js`.

5.4 Analytic Sine and Exponential Julia

For eligible Julia recipes, c is shared and f is analytic. Use a quadratic prefix $\delta_n = A_n u + B_n u^2 + O(u^3)$ with $A_0 = R$, $B_0 = 0$, and $|u| \le 1$. Expanding and collecting degrees yields

$$\delta_{n+1} = f'(Z_n)\delta_n + \tfrac{1}{2} f''(Z_n)\delta_n^2 + O(|\delta_n|^3), \tag{5.33}$$

$$A_{n+1} = f'(Z_n)A_n, \tag{5.34}$$

$$B_{n+1} = f'(Z_n)B_n + \tfrac{1}{2} f''(Z_n)A_n^2. \tag{5.35}$$

Here $f'' = -f$ for sine and $f'' = f$ for exp. Put $s_n = |A_n| + |B_n|$, $D_n = |f'(Z_n)|$ and $L_n = 2(|f(Z_n)| + D_n + 1)$. The implemented error estimate is

$$\begin{aligned} E_{n+1} = {} & (D_n + L_n(s_n + E_n))E_n \\ & + L_n\big[(2|A_n| + |B_n|)|B_n| + s_n^3\big] + \text{rounding allowance}. \end{aligned} \tag{5.36}$$

The mixed expression counts the omitted degrees three and four in $(A_n u + B_n u^2)^2$; the cubic term accounts for the local Taylor remainder at the scale used by the code.

The 96×96 reference tiles require at least eight prefix steps, amplitude below 10^{-4}, relative estimated error at most 10^{-10}, and separation from the bailout. At $|\delta|^2 \le 10^{-8}$ the local delta uses a cubic series for sine or exponential and a quadratic correction for cosine. Otherwise it uses the identities above until $|\delta| > 0.1$ or the trace ends. It can then continue in Float64 from the reconstructed state. An uncertain result triggers a reference shifted by 48 pixels, and either a non-ambiguous retry or agreement within 10^{-4} in the same class can be accepted. Remaining cases use direct evaluation. Agreement is not a proof that two approximations are both correct. See `analytic-julia.js`.

5.5 Block acceptance and raster reuse

From $x \ge 7$ on rasters of at least 10,000 pixels, a worker may try one orbit for a small block, up to 16 columns and four of its rows. The enclosing disk radius is half the distance between the extreme sample centers. For $|h| \le r$, integrating the derivative gives

$$(z+h)^k - z^k = \int_0^1 k(z+th)^{k-1} h \, dt, \tag{5.37}$$

$$|(z+h)^k - z^k| \le k(|z| + r)^{k-1} r, \tag{5.38}$$

$$|(z+h)^2 - z^2| \le (2|z| + r)r. \tag{5.39}$$

The last line follows directly by factoring the square. Axis folds are nonexpansive. Warp's additional derivative has magnitude at most 1.05 per component. For exponential and sine, the bounds used are based on

$$|e^{z+h} - e^z| \le e^{\Re z + r} r, \tag{5.40}$$

$$|\cos(a + \mathrm{i}b)|^2 = \cos^2 a + \sinh^2 b, \tag{5.41}$$

$$|\sin(z + h) - \sin z| \le |\cos z| r + \tfrac{1}{2} \cosh(|b| + r) r^2. \tag{5.42}$$

The sine remainder follows from the integral remainder with $f'' = -\sin$ and $|\sin(a + \mathrm{i}b)| \le \cosh b$. The Ring and Cross distances are 1-Lipschitz, so lower and upper bounds on their minima come from the per-step intervals. A block is filled only when the implementation's decision and scalar error tests pass. Otherwise it tries eligible fixed-point block evaluation or samples individual pixels. See `orbit-bounds.js` and `worker.js`.

The field cache separates scalar orbit results from RGBA. Recoloring is $O(WH)$ without iterating the orbits. Shallow raster planning can share a frame with fullscreen; Studio and deep views render at viewport size to avoid invisible work. The cache retains up to three frames subject to a 96 MiB budget (it keeps at least one frame). Progressive passes use strides $6, 2, 1$ shallow and $12, 4, 1$ deep. The latter add approximately $1/12^2 + 1/4^2 = 6.94\%$ to the sample count before dimension rounding. Up to four workers interleave rows; their reference caches are separate.

When panning by an integer grid displacement with overlap, the number of newly evaluated positions follows by subtracting the intersection of two rectangles:

$$A_{\text{new}} = WH - (W - |d_x|)(H - |d_y|) \tag{5.43}$$

$$= H|d_x| + W|d_y| - |d_x d_y|. \tag{5.44}$$

Reuse requires the same raster, span, recipe and iteration budget. The old grid preserves coordinates for copied samples and carries the fractional drag remainder. Large shifts or missing overlap require a full render. A deep view also recenters after a cumulative shift beyond four raster widths. See `render-cache.js`.

5.6 Fixed-point transcendental evaluation

The trigonometric kernel uses a quadrant reduction based on Machin's identity and a split grid:

$$\pi/4 = 4 \arctan(1/5) - \arctan(1/239), \tag{5.45}$$

$$r = j\, 2^{-20} + \delta, \qquad 0 \le \delta < 2^{-20}, \tag{5.46}$$

$$j = 2^{10} h + \ell, \qquad j\, 2^{-20} = h\, 2^{-10} + \ell\, 2^{-20}. \tag{5.47}$$

The two anchor tables are built on demand. Angle addition composes their sine and cosine into a direct-mapped cache of 4,096 entries. Horner evaluation uses the Maclaurin correction polynomials

$$\sin \delta = \delta + \delta P_{\sin}(\delta^2), \tag{5.48}$$

$$\cos \delta = 1 + P_{\cos}(\delta^2). \tag{5.49}$$

Warp requests only one reconstructed component. For exp, range reduction and guard precision are

$$\ln 2 = 2\,\mathrm{atanh}(1/3), \tag{5.50}$$

$$t = k\ln 2 + r, \qquad 0 \le r < \ln 2, \tag{5.51}$$

$$e^t = 2^k e^r, \tag{5.52}$$

$$b_e = b + 24 + \lceil \log_2 b \rceil. \tag{5.53}$$

Exp uses the same residual grid, composes anchors multiplicatively, and applies 2^k with a shift. Complex sine derives $\cosh b$ and $\sinh b$ from $e^{|b|}$ and one reciprocal. Related table-based reduction and Taylor evaluation at multiple precision were described by Johansson [19]; this implementation uses its own grid, cache, and operation mix.

Other savings in `precise-formulas.js` include precomputed Drift tables, no initial square for direct Sine/Exp unless tracing, a shift for Phoenix $m = -1/2$, and early termination when a deterministic fixed-point state repeats at a checkpoint (single autonomous step, without Phoenix). Ring retains the largest squared radius below ρ^2 and the smallest above. By monotonicity on each side it needs at most two integer square roots, after the orbit rather than on every iteration. The integer root update is $r' = \lfloor (r + \lfloor v/r \rfloor)/2 \rfloor$. Lyapunov keeps the state update fixed-point, but normally forms the derivative factor in Float64 and batches at most 16 factors before taking a log; it retries with a fixed-point derivative near zero. Newton uses $|z^2|^2 = |z|^4$ on the unrounded square and avoids a generic complex numerator for a unit target.

6 Arithmetic work model

6.1 Assumptions and units

Fix a 1010×751 raster, so $M = 758{,}510$, and a budget $N = 200$ per orbit. Count every iteration as a completed, direct high-precision step. Cubic Two-Seed has two orbits; Lyapunov adds 24 warm-up steps. Auto iterations, early termination, region acceptance, and perturbation are excluded. Assume warm transcendental anchor caches.

For nominal operand widths a, b and a numerator of n bits divided by a d-bit denominator, use the schoolbook weights

$$M_0(a, b) = ab, \tag{6.1}$$

$$D_0(n, d) = d(n - d), \qquad n > d. \tag{6.2}$$

They count a rectangular product of digits and roughly $n - d$ quotient positions, respectively. They are not a claim about the algorithms or constants of a JavaScript engine [18]. Ignore signs, integer parts, leading zeros, linear operations, small integer constants, range reduction with a small quotient, coordinate setup, cold caches, and final coloring. Ring's final roots and the near-zero Lyapunov second pass are excluded. Large intermediate magnitudes or small residuals can change actual operand sizes.

6.2 Primitive work and transcendental degree

Three products form the complex square and two more form the squared escape norm. Each subsequent complex multiplication adds four. Hence for explicit powers and their conjugates,

$$n_{\mathrm{mul}}(k) = 3 + 4(k-2) + 2, \tag{6.3}$$

$$C_{\mathrm{power}}(b,k) = [5 + 4(k-2)]b^2. \tag{6.4}$$

Folds add sign operations, not full-width products. The division weights used below are

$$D_0(2b,b) = b^2, \qquad D_0(3b,2b) = 2b^2, \qquad D_0(5b,4b) = 4b^2.$$

Table 2: Modeled work per completed direct iteration.

Path or primitive	Operation inventory	Work
Classic Mandelbrot, Julia, Ship	Five products; two decimal rescaling divisions	$7b_d^2,\ b_d = d\log_2 10$
Powers and folds	$3 + 4(k-2)$ products for the map; two for the norm	$[5 + 4(k-2)]b^2$
Lambda	Five base products; four for the parameter multiplication	$9b^2$
Cubic	Five base; four for $z^2 z$; two for $0.9z$	$11b^2$ per seed
Phoenix	Five base; two for generic memory; a shift for $m = -1/2$	$5b^2$ default, $7b^2$ generic
Newton / Nova	Five base; a $2b$-bit square; two $5b/4b$ divisions; two convergence products for a single step	$19b^2$ for a single unit-target step; add $4b^2$ for a generic Newton target; subtract $2b^2$ in multistep recipes
Magnet	Five base; two for the denominator norm; four for the numerator; two $3b/2b$ divisions; three for quotient squaring; two for single-step convergence	$20b^2$ single-step, $18b^2$ multistep
Ring / Cross	Reuse the orbit's products; extra comparisons and possible roots after the loop	Same modeled loop cost
Lyapunov	Two products per state step; derivative usually Float64	$2b^2$, over $N + 24$ steps
Warp	Five base; two scaling products; two real sines	$7b^2 + 2S_b$
Exponential	Two output products; two for the norm; sin/cos and exp	$4b^2 + P_b + X_b$
Sine	As for exponential, plus one reciprocal	$5b^2 + P_b + X_b$

The rational widths follow from their fixed-point scales. With $U = 2^b$, the unrounded Newton square has scale U^2. The denominator is $(\widehat{a}^2 + \widehat{b}^2)^2$ at U^4. The unit-target numerator is shifted by $3b$, from U^2 to U^5, so the ratio is scaled by U. With a generic target, the complex numerator starts at U^3 and is shifted by $2b$. In Magnet, the squared denominator and raw complex numerator both start at U^2; the numerator is shifted by b.

For an argument residual smaller than 2^{-20}, the maximum size of a degree-n term in fixed-point units is $2^{v-20n}/n!$. Define the last retained degree by

$$K(v) = \max\{n \geq 1 : 20n + \log_2(n!) \leq v\}, \tag{6.5}$$

$$K_t = K(b), \qquad K_e = K(b_e). \tag{6.6}$$

The kernel retains at least the linear term and implements the cutoff with the integer test $(2^v \gg 20n) \geq n!$ for $n \geq 2$. This test is a truncation rule for a term, not a certificate of total rounding error.

There are $K_t - 1$ correction coefficients across the sine and cosine polynomials. Horner uses one product per correction coefficient, including the final factor δ^2. Add one product for δ^2, one for the remaining sine factor δ, and two per reconstructed component. Exp uses K_e Horner products and one anchor product. Thus, on an anchor cache hit,

$$S_b = [(K_t - 1) + 1 + 1 + 2]b^2 = (K_t + 3)b^2, \tag{6.7}$$

$$P_b = [(K_t - 1) + 1 + 1 + 4]b^2 = (K_t + 5)b^2, \tag{6.8}$$

$$X_b = (K_e + 1)b_e^2. \tag{6.9}$$

An anchor miss costs four additional composition products for trig or one for exp; cold component tables add series construction. Stirling's formula explains the eventual model scaling:

$$20n + \log_2(n!) = n(20 + \log_2 n - \log_2 e) + O(\log n), \tag{6.10}$$

$$K(v) = \Theta(v/\log v), \tag{6.11}$$

$$C_{\text{residual}}(b) = \Theta(b^3/\log b). \tag{6.12}$$

The last line applies to this residual evaluation and these schoolbook weights; it excludes table construction and does not describe actual BigInt timing.

6.3 Recipe totals and worked calculations

Let $A_f(k)$ count occurrences of step f among the first k entries of the expanded word S. Count distinct step settings separately when their work differs. For a word of length L and q seeds, full cycles and the final prefix give

$$n_f(N) = \lfloor N/L \rfloor A_f(L) + A_f(N \bmod L), \tag{6.13}$$

$$C(x) = q \sum_f n_f(N) C_f(b(x, W)), \tag{6.14}$$

$$G(x) = 100 \left[\frac{C(x+1)}{C(x)} - 1 \right]. \tag{6.15}$$

Multiply $C(x)$ by M for a fully direct final frame. Use $N+24$ for Lyapunov. For a fixed operation mix in binary, with $A = x + \log_{10} W + 32$,

$$\frac{C(x+1)}{C(x)} = \left(\frac{b(x+1, W)}{b(x, W)} \right)^2, \tag{6.16}$$

$$\frac{G(x)}{100} \approx \left(\frac{A+1}{A} \right)^2 - 1 = \frac{2}{A} + \frac{1}{A^2} \quad \text{(ignoring the bit ceiling)}. \tag{6.17}$$

For the classic path, start at $d = 40$ and increase x by integer decades with $s = 3.3\,10^{-x}$:

$$d(x) = 40 + 20\max\{0, \lceil (x + 24 - 40)/20 \rceil\}, \tag{6.18}$$

$$\frac{C(17)}{C(16)} = (60/40)^2 = 2.25. \tag{6.19}$$

For integer x, the scaled span has at least 25 digits when $d - x \geq 24$, which yields the first line. A manually entered or previously deeper view can retain a larger d.

Table 3: Precision and retained degrees at selected magnifications.

x	Classic d	Studio b	Exp b_e	K_t	K_e
12	40	157	189	7	8
13	40	160	192	7	8
14	40	163	195	7	8
15	40	167	199	7	9
16	40	170	202	7	9
17	60	173	205	7	9
18	60	177	209	8	9
40	80	250	282	11	12
41	80	253	285	11	12
100	140	449	482	19	20
101	140	452	485	19	20

Quadratic Studio. The common baseline and two growth factors are

$$C_{z^2}(12) = 200 \cdot 5 \cdot 157^2 = 24{,}649{,}000,$$
$$C_{z^2}(13)/C_{z^2}(12) = (160/157)^2 \approx 1.03858,$$
$$C_{z^2}(100)/C_{z^2}(12) = (449/157)^2 \approx 8.17887.$$

A seventh power uses 25 products rather than 5 per step, but has the same relative growth.

Warp. At the two endpoints, the direct work per iteration is

$$C_{\mathrm{Warp}}(12)/N = [7 + 2(7 + 3)]157^2 = 27 \cdot 157^2,$$
$$C_{\mathrm{Warp}}(100)/N = [7 + 2(19 + 3)]449^2 = 51 \cdot 449^2,$$
$$C_{\mathrm{Warp}}(100)/C_{\mathrm{Warp}}(12) = (51/27)(449/157)^2 \approx 15.45.$$

The degree change from $x = 17$ to 18 gives $(29/27)(177/173)^2 \approx 1.1243$.

Mixed Drift z^3 and Warp. There are 50 cubic steps and 150 Warp steps at $N = 200$:

$$\begin{aligned} n_{\sin} &= 2 \cdot 150 = 300 \quad \text{per direct pixel,} \\ C_{\mathrm{mix}}(12) &= (50 \cdot 9 + 150 \cdot 27)157^2 \\ &= 110{,}920{,}500, \\ MC_{\mathrm{mix}}(12) &= 84{,}134{,}308{,}455{,}000, \\ Mn_{\sin} &= 227{,}553{,}000, \\ C_{\mathrm{mix}}(100)/C_{\mathrm{mix}}(12) &= (8{,}100/4{,}500)(449/157)^2 \approx 14.72. \end{aligned}$$

The frame totals are conditional on every pixel taking the full direct route. For a separate scenario with a mean of 117 Warp steps per direct pixel and a 320×200 entirely direct raster, the same calculation gives $2 \cdot 117 \cdot 320 \cdot 200 = 14{,}976{,}000$ sines. That is a conditional extrapolation, not a frame-wide counter.

Sine Julia and Exponential Julia. For Sine Julia, substitute b_e as well as b:

$$\begin{aligned} C_{\mathrm{Sine}}(12)/N &= [5 + (7 + 5)]157^2 + (8 + 1)189^2 = 740{,}522, \\ C_{\mathrm{Sine}}(100)/N &= [5 + (19 + 5)]449^2 + (20 + 1)482^2 = 10{,}725{,}233, \\ C_{\mathrm{Sine}}(100)/C_{\mathrm{Sine}}(12) &\approx 14.48334. \end{aligned}$$

Exponential replaces the leading 5 with 4 by omitting the reciprocal, yielding a ratio of approximately 14.70042.

7 Results and reproducibility

7.1 Modeled results by family

Table 4 groups presets with the same sampled decade growth. Appendix A gives all 52 individual presets and the running recipe. All numbers are work ratios at fixed budgets for the direct path; zero growth for a classic decade means d did not change in that decade.

Table 4: Percent growth for selected zoom decades and the total work ratio.

Group	$G(12)$	$G(16)$	$G(40)$	$G(100)$	$C(100)/C(12)$
Six classic presets	0%	125.00%	0%	0%	12.25
Forty Studio presets with a fixed operation mix	3.86%	3.56%	2.41%	1.34%	8.18
Warp	3.86%	3.56%	2.41%	1.34%	15.45
Singularity / Drift Singularity	3.86%	3.56%	2.41%	1.34%	12.98 / 14.31
Kaleidoscope	3.86%	3.56%	2.41%	1.34%	11.86
Exponential Julia	3.56%	3.29%	2.29%	1.30%	14.70
Sine Julia	3.57%	3.30%	2.29%	1.30%	14.48

Group	$G(12)$	$G(16)$	$G(40)$	$G(100)$	$C(100)/C(12)$
Mixed Drift z^3 and Warp	3.86%	3.56%	2.41%	1.34%	14.72

7.2 Archived frame latency

The measured frame includes setup, reference traces, block attempts, per-pixel retries, direct fallbacks, and output. An accounting model for one worker is

$$T_{\text{worker}} \simeq T_{\text{setup}} + \sum_t T_{\text{ref},t} + \sum_B T_{\text{block},B} + \sum_i T_{\text{try},i} + \sum_{i \in F} \sum_{n < N_i} t_{\text{direct},f_n}(b) + T_{\text{transfer}}. \tag{7.1}$$

Here F is the set of direct fallback pixels, N_i is actual iteration work, and t is a time, unlike C_f. The slowest worker in each pass, assembly and coloring affect visible latency. When enabled, Auto iterations additionally increase the budget:

$$N_{\text{auto}} = 100 \ \text{round}\left(\frac{600 + 170 \max(0, x - 2)}{100}\right),$$
$$N = \max(N_{\text{chosen}}, \min(20{,}000, N_{\text{auto}})).$$

Table 5 uses archived 1010×751 views with four workers and one run per view. The renderer ran in Node and timings include worker startup, progressive passes, assembly and RGBA, but not page load or browser painting. The raw records retain exact state: Warp and Mandelbrot. The Warp rows share a center; the mini-Mandelbrot rows use different centers selected for visible structure.

The visibility filter bins each color channel into eight levels. A bin is significant at a share of at least 1%, and a tile in a 12×8 grid is varied if it has at least three bins. An edge is a horizontal or vertical neighbor pair with a channel difference of at least 24. Selection required at least three significant bins, at most 90% in the largest bin, at least 24 varied tiles, and an edge share of at least 0.2%. Background means the exact RGB value $(10, 14, 25)$. Neither edge share nor background measures fallback rate or iteration count.

Table 5: Archived frame timings and image detail metrics. Each row represents one run.

View	Zoom x	Budget	Preview (s)	Final (s)	Background	Edges
Mixed Warp (wider view)	11.00000	200	0.143	2.374	7.20%	29.79%
Mixed Warp (study view)	11.44672	200	1.220	102.087	11.79%	32.34%
Mixed Warp (narrower view)	14.00000	200	1.127	97.558	5.44%	11.85%
Mandelbrot (same center)	18.00000	3000	0.123	3.813	0.00%	37.75%
Mandelbrot (same center)	20.50389	3000	0.096	2.346	0.00%	17.67%

Continued on the next page

View	Zoom x	Budget	Preview (s)	Final (s)	Background	Edges
Mini-Mandelbrot	11.52169	3000	0.084	1.502	15.87%	14.95%
Mini-Mandelbrot	17.54229	3000	0.087	1.406	15.99%	18.12%
Mini-Mandelbrot	40.11954	3000	0.088	1.868	16.47%	23.47%
Mini-Mandelbrot	99.92803	3000	0.128	4.149	18.78%	32.62%

The wider Warp frame lies just below the strict $x > 11$ threshold in the stored state; the additional Float64 resolution test is false. The next row enters the deep route. The roughly 43× change cannot be attributed to a few extra bits. The $x = 14$ Warp frame then finishes earlier than the $x \approx 11.45$ frame. These are observations for the selected views, not a fitted latency law.

7.3 Historical optimization comparisons

The optimization archive records 320×200, $N = 200$, one-worker comparisons. A row uses the same view before and after, but the archive does not pin a common pair of revisions across all rows. Some controls already included trig improvements. These are descriptive development measurements, not single-optimization ablations or a comparison against competing engines.

Table 6: Historical before–after timings and direct evaluations in the final pass.

Case	Before (s)	After (s)	Direct before	Direct after
Cubic Two-Seed	21.963	1.877	63,875	274
Phoenix	8.312	5.461	64,000	32,670
Magnet	29.401	20.176	64,000	37,352
Nova	27.190	22.722	64,000	54,172
Sine (parameter plane)	58.976	12.576	55,319	12,184
Ring	4.834	3.348	35,157	35,157
Lyapunov	5.147	3.754	64,000	64,000
Lyapunov (deep)	6.258	4.799	63,999	63,999

The archive records no switch between $V = N$ and $V \neq N$ in these comparisons. The maximum scalar difference was below 1.3×10^{-11}. This is not bitwise equivalence in general. The archive also records a negative trial: a reciprocal plus correction preserved the field for Nova and Magnet in the tests but did not reduce runtime; it was not retained.

7.4 Reproduction and limitations

The following commands reproduce the operation tables, rerun selected views on the current engine, and check the implementation:

```
node scripts/fractal-cost-model.mjs --csv
node scripts/fractal-cost-model.mjs --decades
```

```
node scripts/benchmark-paper.mjs mini12 3
node scripts/check.mjs
```

The cost script reads the live catalog and checks classic precision against `ensurePrecision`. It emits 53 recipe summaries and 4,823 recipe–depth records across 10^{11} to 10^{101}. The benchmark script reads the saved views and records all repetitions and the current environment. The tests cover direct versus accelerated fields, shuffled recipes, very deep coordinates, pan strips, and palette/cache behavior. The accepted numerical tolerances are route dependent. The direct evaluator is itself quantized.

Stronger performance evidence would pin all baselines, isolate each optimization, collect fallback fractions, orbit-length histograms and anchor-cache misses, stratify scenes by texture and numerical conditioning, and report dispersion over repetitions. Numerical validation should also compare to an independent multiprecision or ball-arithmetic evaluator and distinguish class errors from scalar differences. Nothing in the existing archive establishes superiority to an external renderer.

8 Prior art and contribution assessment

8.1 Search scope

The accompanying search and evidence log records queries, primary sources, and unresolved comparisons. The public search was conducted on 12 November 2026 and covered papers, preprints, publisher pages, author notes, software manuals, and public renderer repositories. Searches combined numerical methods with fractal names and alternate spellings. A match was treated as method-level evidence only if the source described the relevant evaluation path, rather than merely listing a fractal in its catalog. This was a broad public prior-art search, not an exhaustive priority or patent determination. Unindexed forum archives, proprietary implementations, and inaccessible source versions remain gaps.

8.2 What has clear precedent

Perturbation, prefixes, rebasing, and browser deployment. Martin and Khashin describe shared reference orbits for Mandelbrot; Heiland-Allen documents rebasing and block linear approximations [7, 8, 9]. Fraktaler 3 explicitly loops editable formula lines, supports folds and a small opcode language, and implements perturbation and BLA for hybrid fractals [10]. Kalles Fraktaler is another concrete precedent for folded and hybrid perturbation [11]. Ultra Fractal gives authors an explicit interface for formula-specific perturbation equations [12]. Other implementations use hybrid bytecode, matrix linear approximations, and a double-double tier [13]. Browser-based high-precision reference orbits combined with GPU perturbation are also documented. In particular, Deck describes exact scaled recurrences for Phoenix, Nova, Magnet, and other maps, and extended-exponent pixel differences [14]. Mandelbrot Scout independently documents Phoenix perturbation and the difficulty of reusing ordinary Mandelbrot rebasing for delayed state [15]. Fractalia cannot claim priority for hybrid formulas, formula-aware perturbation, linear skipping, browser deep zoom, or the application of reference differences to those named families.

Arithmetic and region shortcuts. Dekker's multilength floating-point arithmetic predates the double-double recovery path [16]. Johansson describes multiple-precision sine, cosine, and exp using table-based argument reduction and Taylor evaluation [19]. Multiprecision ball arithmetic

provides a stronger enclosure framework than the floating-point margins used here [20]. Taking norms after composing linear responses is standard linear algebra; the broader problem of losing correlations in enclosure propagation is also well known in validated numerics [21]. Mariani–Silver illustrates prior region reuse based on rectangular borders [22]. Fractalia's block predicate instead propagates a local disk and checks the observable, and its Float64 margins do not constitute certified intervals. Orbit traps also predate this implementation [23].

8.3 What this report contributes

A concrete, inspectable integration. Fractalia applies a single recipe interface to folded powers, transcendental steps, rational steps, and delayed Phoenix state, with a separate mode for the forced Lyapunov observable. Julia and Drift are per-step settings. The renderer adapts its metadata, tile policy, prefix eligibility, error tests, and fallback to those settings. That is an engineering contribution about this implementation, not proof that no other renderer supports the same combination.

An engine-specific result. The full-budget operation inventory and decade-by-decade ratios are computed from Fractalia's code, including its precision transitions and residual polynomial cutoffs. The 52-preset table, the mixed Warp decomposition, and the archived frame comparisons are reproducible artifacts or documented observations of this project. They are not new theorems about fractals or comparative speed claims.

Candidates for further investigation. Fractall's Lyapunov implementation already has a direct multiprecision evaluator. Its hybrid compiler excludes Lyapunov and several special orbit families from that bytecode route; this says nothing about every separate route in the project. The most specific candidates here are the eight-step cancellation-aware Phoenix acceptance policy, the local pole-aware derivative bounds for rational steps, and a Lyapunov reference kernel whose decision includes uncertainty in the observable's sign. We found neighboring techniques and renderers listing these families, but did not verify an equivalent implementation of those exact *acceptance* policies in the sources reviewed. That limited negative result does not establish priority: the identities follow from standard algebra, and catalogs are not complete documentation of numerical paths. A stronger paper would isolate each policy against a well-defined baseline, validate against an independent reference, and compare to the closest available implementations on matched scenes.

8.4 Documentation and materials

Source code, state fixtures, arithmetic scripts, and historical raw results are in the repository. This report is a technical draft; it does not assign personal authorship or imply submission approval. The evidence review and editorial structure used the Scientific Writing skill as guidance [24]; the skill is not evidence for any mathematical or performance claim.

References

[1] G. Julia. Mémoire sur l'itération des fonctions rationnelles. *Journal de Mathématiques Pures et Appliquées*, 1:47–245, 1918.

[2] B. B. Mandelbrot. Fractal aspects of the iteration of $z \mapsto \lambda z(1-z)$ for complex λ and z. *Annals of the New York Academy of Sciences*, 357:249–259, 1980.

[3] M. Michelitsch and O. E. Rössler. The “burning ship” and its quasi-Julia sets. *Computers & Graphics*, 16(4):435–438, 1992.

[4] J. Hubbard, D. Schleicher, and S. Sutherland. How to find all roots of complex polynomials by Newton’s method. *Inventiones Mathematicae*, 146:1–33, 2001.

[5] L. Vepstas. Renormalizing the Mandelbrot Escape. Technical note, 1997.

[6] M. Markus and B. Hess. Lyapunov exponents of the logistic map with periodic forcing. *Computers & Graphics*, 13(4):553–558, 1989.

[7] K. I. Martin. SuperFractalThing Maths. Archived technical note, 2013.

[8] S. Khashin. Fast calculation of the Mandelbrot set with infinite resolution. Preprint, 2016.

[9] C. Heiland-Allen. Deep zoom theory and practice. Technical note, 2021.

[10] C. Heiland-Allen. Fraktaler 3.1 manual. See Formula Window and the algorithm notes.

[11] K. Runmo and contributors. Kalles Fraktaler 2+ manual. See the hybrid formula and perturbation documentation.

[12] Ultra Fractal. Perturbation equations. Formula-writing manual.

[13] Fractall contributors. Fractall: Fractal Explorer in Rust. Project documentation, accessed 12 November 2026.

[14] C. Deck. Fractal Explorer: Deep Zoom in the Browser. Author’s technical note, 2026.

[15] Mandelbrot Scout contributors. Mandelbrot Scout, version 3.4 alternative-formula notes. Public project documentation.

[16] T. J. Dekker. A floating-point technique for extending the available precision. *Numerische Mathematik*, 18:224–242, 1971.

[17] NIST. *Digital Library of Mathematical Functions*. Exponential and logarithm, trigonometric series, and hyperbolic functions.

[18] R. P. Brent and P. Zimmermann. Modern Computer Arithmetic. Version 0.5.1, 2010.

[19] F. Johansson. Efficient implementation of elementary functions in the medium-precision range. Preprint revised 2015.

[20] F. Johansson. Arb: Efficient Arbitrary-Precision Midpoint-Radius Interval Arithmetic. *IEEE Transactions on Computers*, 2017.

[21] M. Berz and K. Makino. New approaches for the validation of transfer maps using remainder-enhanced differential algebra. *Nuclear Instruments and Methods in Physics Research A*, 519:53–62, 2004.

[22] R. Mariani. The Mariani-Silver Algorithm for Drawing the Mandelbrot Set. Author’s retrospective.

[23] Ultra Fractal. Orbit Traps. Standard coloring manual.

[24] T. Kassis, V. Agarwal, Y. He, D. Patel, and A. M. Brueckner. Scientific Agent Skills: A Library of Procedural Knowledge for Research Agents. Preprint, 2026.

A Results for every preset

The first three rows are the original presets, followed by the Studio catalog and the running recipe. Formula constants, seeds, and modifiers are available in `catalog.js` and `app.js`. The work at 10^{12} is relative to the quadratic Studio baseline. Calls are reported as single-component sine / sine–cosine pairs / exponential, per pixel at the full modeled budget.

Table 7: Modeled direct work for every preset and the mixed study recipe.

Preset	Path	Recipe	Steps per pixel	Relative work at 10^{12}	$\frac{C(100)}{C(12)}$	Calls per pixel
Seahorse Valley	Classic	mandelbrot	200	1.00	12.25	0/0/0
Julia's Garden	Classic	julia	200	1.00	12.25	0/0/0
Burning Shores	Classic	ship	200	1.00	12.25	0/0/0
Multibrot z^3	Studio	z3	200	1.80	8.18	0/0/0
Multibrot z^4	Studio	z4	200	2.60	8.18	0/0/0
Multibrot z^5	Studio	z5	200	3.40	8.18	0/0/0
Multibrot z^6	Studio	z6	200	4.20	8.18	0/0/0
Multibrot z^7	Studio	z7	200	5.00	8.18	0/0/0
Julia Dendrite	Classic	z2+Julia	200	1.00	12.25	0/0/0
Julia Snowflake	Studio	z5+Julia	200	3.40	8.18	0/0/0
Tricorn	Studio	tricorn	200	1.00	8.18	0/0/0
Multicorn z^3	Studio	multi3	200	1.80	8.18	0/0/0
Multicorn z^4	Studio	multi4	200	2.60	8.18	0/0/0
Multicorn z^5	Studio	multi5	200	3.40	8.18	0/0/0
Celtic	Studio	celtic	200	1.00	8.18	0/0/0
Buffalo	Studio	buffalo	200	1.00	8.18	0/0/0
Perpendicular Ship	Studio	perpendicular	200	1.00	8.18	0/0/0
Heart	Studio	heart	200	1.00	8.18	0/0/0
Warp	Studio	warp	200	5.40	15.45	400/0/0
Phoenix	Studio	phoenix+Julia	200	1.00	8.18	0/0/0
Lambda	Studio	lambda	200	1.80	8.18	0/0/0
Cubic Two-Seed	Studio	cubic	400	4.40	8.18	0/0/0
Elephant Valley	Classic	z2	200	1.00	12.25	0/0/0
Ship Armada	Classic	ship	200	1.00	12.25	0/0/0
Orbit Rings	Studio	z2	200	1.00	8.18	0/0/0
Orbit Cross	Studio	z2+Julia	200	1.00	8.18	0/0/0
Fractaloid	Studio	z3 → tricorn×2 → z2	200	1.20	8.18	0/0/0
Mirror Ship	Studio	ship×2 → tricorn	200	1.00	8.18	0/0/0
Pulse	Studio	z2×4 → z5	200	1.48	8.18	0/0/0
Singularity	Studio	warp → celtic → z3	200	2.74	12.98	134/0/0

Continued on the next page

Preset	Path	Recipe	Steps per pixel	Relative work at 10^{12}	$\frac{C(100)}{C(12)}$	Calls per pixel
Chainmail	Studio	celtic → buffalo → z2×3	200	1.00	8.18	0/0/0
Kaleidoscope	Studio	z3+Julia → multi5+Julia → warp+Julia	200	3.52	11.86	132/0/0
Drift Julia	Studio	z2+Julia+Drift	200	1.00	8.18	0/0/0
Drift Seahorse	Studio	z2+Julia+Drift	200	1.00	8.18	0/0/0
Drift Tricorn	Studio	tricorn+Julia+Drift	200	1.00	8.18	0/0/0
Drift Cubic	Studio	z3+Julia+Drift	200	1.80	8.18	0/0/0
Drift Dendrite	Studio	z2+Julia+Drift	200	1.00	8.18	0/0/0
Drift Snowflake	Studio	z5+Julia+Drift	200	3.40	8.18	0/0/0
Drift Heart	Studio	heart+Julia+Drift	200	1.00	8.18	0/0/0
Drift Phoenix	Studio	phoenix+Julia+Drift	200	1.00	8.18	0/0/0
Drift Fractaloid	Studio	z3+Julia+Drift → tricorn+Julia	200	1.40	8.18	0/0/0
Drift Singularity	Studio	warp+Julia+Drift → celtic+Julia	200	3.20	14.31	200/0/0
Drift Pulse	Studio	z2+Julia+Drift → z5+Julia	200	2.20	8.18	0/0/0
Drift Multicorn	Studio	multi5+Julia+Drift	200	3.40	8.18	0/0/0
Newton Basins	Studio	newton	200	3.80	8.18	0/0/0
Nova	Studio	nova	200	3.80	8.18	0/0/0
Nova Jellyfish	Studio	nova+Julia	200	3.80	8.18	0/0/0
Magnet	Studio	magnet	200	4.00	8.18	0/0/0
Exponential Julia	Studio	exp+Julia	200	5.81	14.70	0/200/200
Sine Julia	Studio	sine+Julia	200	6.01	14.48	0/200/200
Lyapunov	Studio	AABAB	224	0.45	8.18	0/0/0
Lyapunov Braid	Studio	ABBABAAB	224	0.45	8.18	0/0/0
Drift z^3 → Warp ×3	Studio	z3+Julia+Drift → warp×3	200	4.50	14.72	300/0/0